Data Mining & Analysis on Amazon Datasets

Barmparosos Theofanis

February 28, 2026

Abstract

This report presents a comprehensive data mining and analysis study conducted on large-scale Amazon datasets. The project is divided into two complementary parts. Part 1 mainly focuses on Exploratory Data Analysis (EDA) and Sentiment Analysis to explore user reviews, ratings and the overall structure of the datasets. Part 2 involves text pre-processing, clustering of products to group them into subsets, a recommendation system and frequent pattern mining to uncover association rules and purchasing patterns. The work is focused on searching (mining) for useful and interesting information on large-scale datasets, the characteristics of which we don't know a priori. The Amazon product categories that were used are the following five. Appliances, All_Beauty, Musical_Instruments, Software and Video_Games.

Contents

1 Project Structure and Technical Setup **3**

2 Datasets **3**

3 Part 1: EDA & Sentiment Analysis **3**
- 3.1 Introduction to Part 1 . . . 3
- 3.2 Exploratory Data Analysis . . . 3
- 3.3 Data Pre-Processing . . . 6
- 3.4 Product Performance Analysis . . . 6
- 3.5 Sentiment Analysis Methodology . . . 15
- 3.6 Results: Ratings vs. Sentiment . . . 15

4 Part 2: Product Analysis, Clustering and Market Basket Mining **16**
- 4.1 Introduction to Part 2 . . . 16
- 4.2 Dataset Preparation and Text Pre-Processing . . . 16
- 4.3 Feature Extraction . . . 16
- 4.4 Product Clustering . . . 16
  - 4.4.1 Clustering Methodology . . . 16
  - 4.4.2 Cluster Selection . . . 16
  - 4.4.3 Cluster Interpretation . . . 16
- 4.5 Cluster-Based Recommendation Strategy . . . 16
- 4.6 Association Rule Mining . . . 16
- 4.7 Temporal Pattern Analysis . . . 16

5 Conclusion **17**

1 Project Structure and Technical Setup

This is a project that aims to accomplish an extensive and precise study on large-scale Amazon datasets. All of the code is written in Python (v3.10.12) using Jupyter notebooks and is organized into two main parts, corresponding to the two analytical phases of the study. All required dependencies must be installed before running any of the notebooks with the following command at the root of the repository: `pip install -r requirements.txt` After installation, the notebooks can be executed.

2 Datasets

Datasets used in this project are from Hugging Face and concern Amazon products, including reviews and metadata collected in 2023 by McAuley Lab. Both parts use the same 10 CSV files: two datasets, reviews and metadata, for each product category. The first few code cells in both notebooks download the datasets locally. One could use the textttmax_rows variable to limit the datasets to an arbitrary number of rows, to save time and system resources. Full-size datasets are quite large and take up approximately 6.4GB of space.

3 Part 1: EDA & Sentiment Analysis

3.1 Introduction to Part 1

The objective of this part is to analyze the overall nature of the datasets, discover average prices, average ratings of the different categories, best and worst selling products and the relationship between user ratings and the textual content of reviews. Natural Language Processing (NLP) techniques are utilized to compute sentiment scores and correlate them with the star ratings.

3.2 Exploratory Data Analysis

After downloading, the datasets are used to create pandas DataFrames that can later be manipulated as desired. The main goal is to explore the nature of the datasets.

- At first, the notebook displays the average product price in each category.

Table 1: Average Product Price per Category

Category	**Average Product Price (USD $)**
Appliances	86.4799
Beauty	27.2557
Musical Instruments	127.7064
Software	2.9567
Video Games	45.7173

In order to ensure that the whole dataset consists of valid data, one step of data preprocessing is performed now, before moving on with the rest of the EDA: handling

missing/bad values in the 'price' column. Further details are provided in the Data Pre-Processing section.

- A set of plots show the distribution of ratings in each category within the 5-star range. In other words, each plot shows how many 1-star, 2-star, 3-star, 4-star and 5-star ratings are there in each category's dataset. It becomes evident that across all product categories, the 1-star and 5-star ratings exhibit significantly higher frequencies compared to the intermediate ratings (2–4 stars). This pattern can be explained by the fact that customers are more likely to leave a review when they feel either extremely dissatisfied or fully satisfied with a product. In contrast, moderately satisfied users tend to provide feedback less frequently.
  Overall, an increasing trend is observed in the rating distributions, indicating that positive reviews (4 and 5 stars) outnumber negative ones (1 and 2 stars).

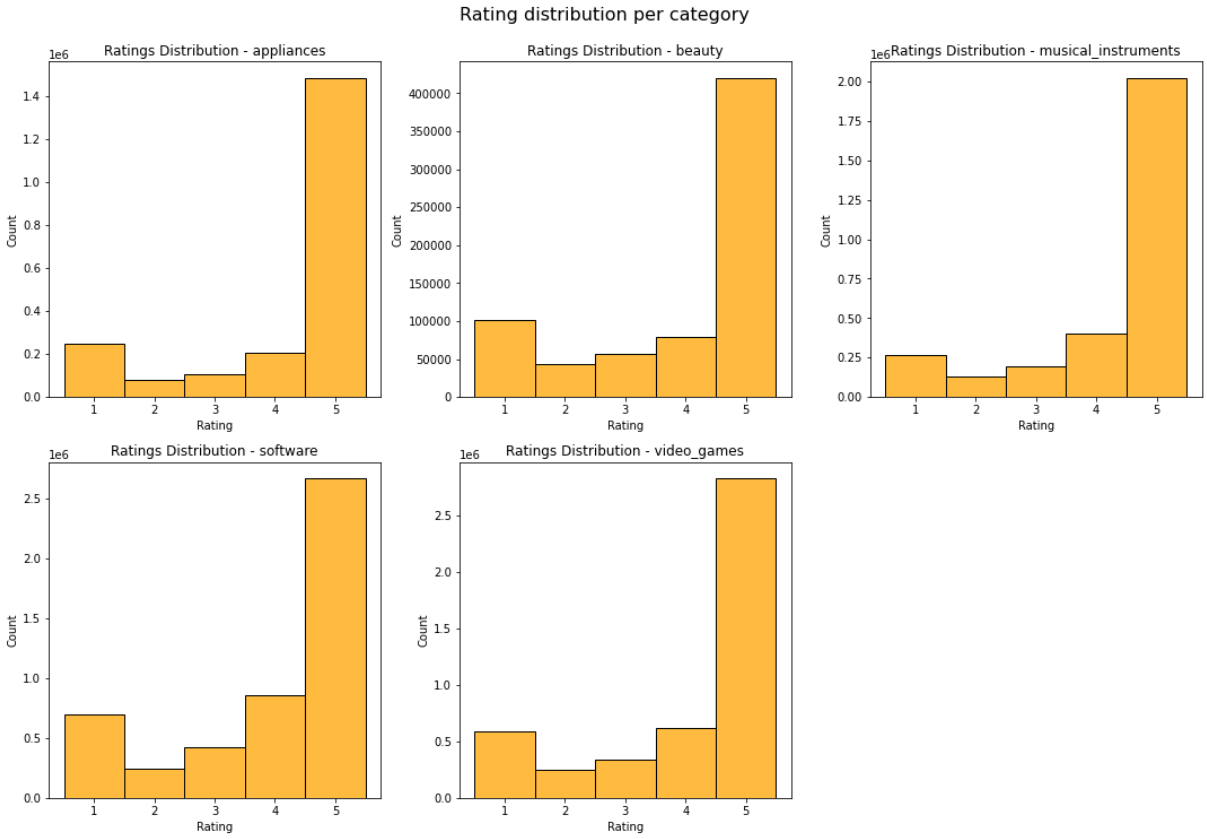
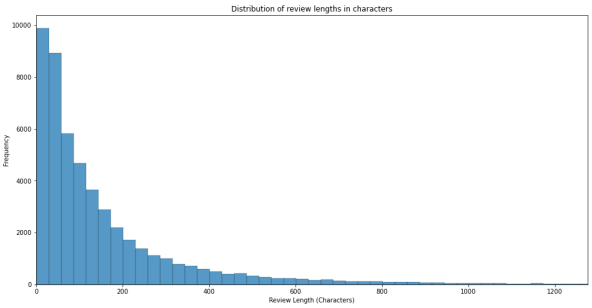
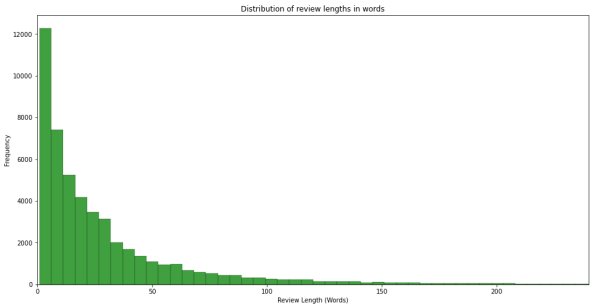


Figure 1: Distribution of Ratings in Each Category

- 2 plots per category follow, showcasing the distribution of review lengths both in words and in characters.

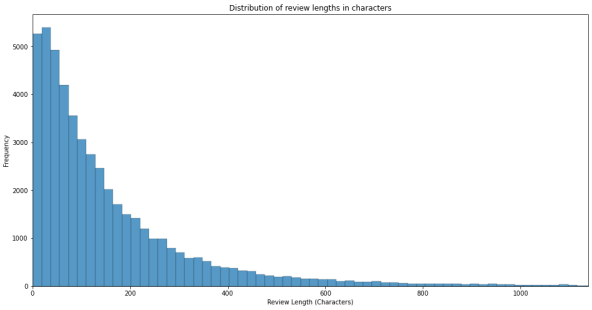


(a) Character Count

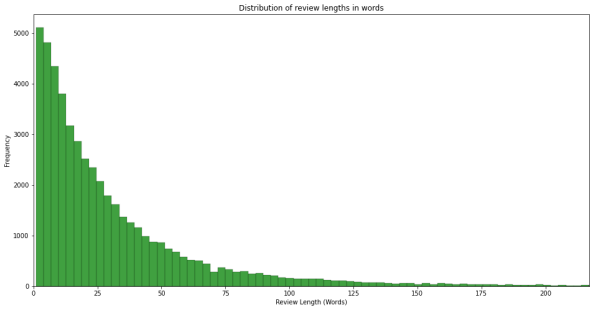


(b) Word Count

Figure 2: Review Length Distribution for Appliances

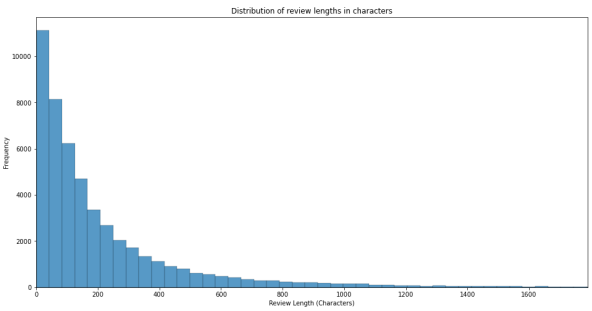


(a) Character Count

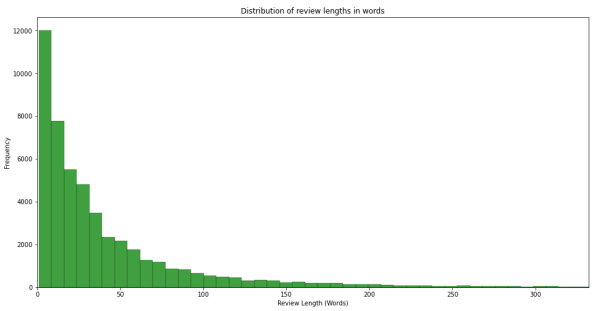


(b) Word Count

Figure 3: Review Length Distribution for Beauty

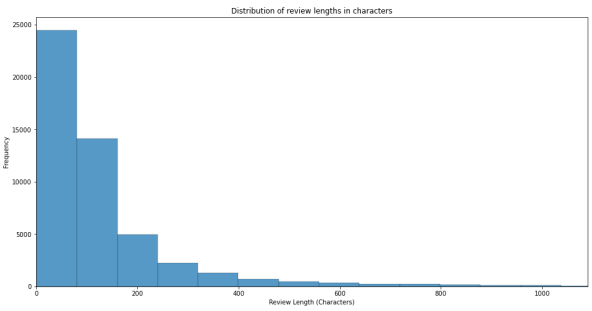


(a) Character Count

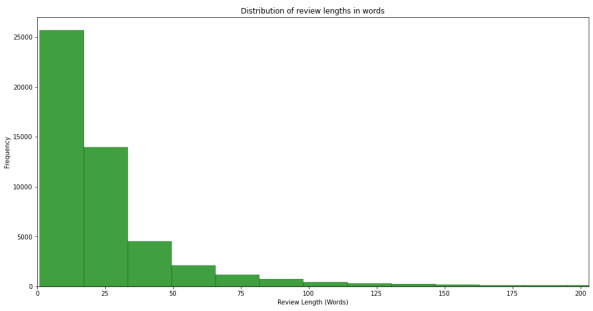


(b) Word Count

Figure 4: Review Length Distribution for Musical Instruments

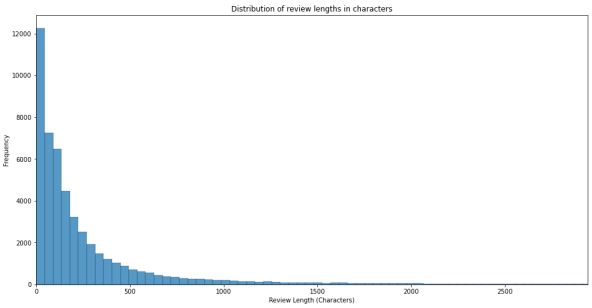


(a) Character Count

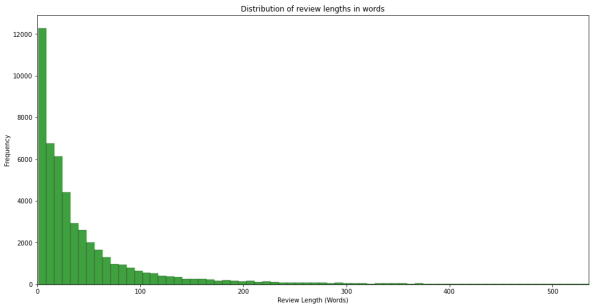


(b) Word Count

Figure 5: Review Length Distribution for Software



(a) Character Count



(b) Word Count

Figure 6: Review Length Distribution for Video Games

- Average product rating is displayed for each category.

Table 2: Average Product Rating per Category

Category	**Average Product Rating(0–5)**
Appliances	4.2215
Beauty	3.9602
Musical Instruments	4.2555
Software	3.9350
Video Games	4.0474

3.3 Data Pre-Processing

- All missing price values are replaced with the mean price value.
- All bad values or NULL values are replaced with the mean price value.
- All HTML elements are removed from the text column of the reviewsreviews(e.g., &amp).

After experimentation, it was found that no other pre-processing was necessary for this part.

3.4 Product Performance Analysis

In this section, an attempt is made to find the best selling products and the worst sellers. To be more specific, Best sellers are products with high review count and high ratings, whereas worst-selling products are those with a high review count but quite low rating. Starting off with the worst sellers, the 10 worst products are presented for each category, along with a word cloud plot per category, to display commonly used vocabulary in the reviews of such products.

- Appliances:
    1. B01CBWDY0C: Nespresso Capsules Refillable - Reusable Coffee Pods For Nespresso Cups - OriginalLine Compatible - Pack of 6 (reviews: 938, avg rating: 3.00)

2. B081K5T5HB: Deflecto Magnetic Vent Cover, For Sidewall and Floor Vents, 5” x 12”, 3 Pack (MVCX512),White (reviews: 896, avg rating: 2.51)
3. B0000CF4DJ: Harold Coffee Tools Filter Separator (reviews: 749, avg rating: 2.87)
4. B08WH5XL86: RCA RFRF450-AMZ, 5.1 Cubic Foot Chest, Deep Freezer Cold Storage for Food, White (reviews: 691, avg rating: 2.77)
5. B09JY77GCY: GE Profile Opal — Countertop Nugget Ice Maker with Side Tank — Portable Ice Machine Makes up to 24 lbs. of Ice Per Day — Stainless Steel Finish (reviews: 675, avg rating: 2.85)
6. B09B18P5RG: GE Profile Opal — Countertop Nugget Ice Maker — Portable Ice Machine Complete with Bluetooth Connectivity — Smart Home Kitchen Essentials — Stainless Steel Finish — Up to 24 lbs. of Ice Per Day (reviews: 501, avg rating: 2.78)
7. B00GN92KQ4: Quirky Egg Minder Wink App Enabled Smart Egg Tray, PEGGM-WH01 (reviews: 500, avg rating: 2.59)
8. B07QPBTR4B: AquaPlumb RKG Gas Range Knob- 5PC/Card (reviews: 403, avg rating: 2.35)
9. B07S61G1RQ: Jetec 4 Pack Black Control Knobs Replacement with 12 Adapters for Oven/Stove/Range (reviews: 395, avg rating: 2.81)
10. B08MMJ8RM7: RCA RFRF454-BLACK 5.1 Cubic Feet Chest Freezer, 5 cu ft, Black (reviews: 379, avg rating: 2.93)

- Beauty:

1. B01195J43I: DASKY 29W Digital Anti Static Ceramic Hair Straightener Heating Detangling Hair Brush Paddle Brush For Faster Straightening Styling Massage Straightening Iron (Pink) (reviews: 502, avg rating: 2.84)
2. B082RD77G5: Purple Shampoo and Conditioner Set - Sulfate Free Salon Grade (2 x 16.9 fl oz) - Hydrating Toner - Shimmer Correction for Platinum Blonde, Silver, Light, and Grey Hair (reviews: 478, avg rating: 2.91)

3. B071XQ355Y: Tooletries – Oliver Shower Mirror - Premium Shower Mirror for Shaving, Grooming - 100
4. B01CTQC9J2: CITY 1 PC White Peel Off Liquid Tape Peel Off Base Coat Nail Art Liquid Palisade (reviews: 338, avg rating: 2.78)
5. B07WDJB6BH: Allnice Magnetic Eyelashes Dual Magnets False Eyelashes Reusable 3D No Glue 3 Fake Lashes Extension with Tweezers for Women Makeup Natural Look 8 Pieces / 2 Pairs (reviews: 317, avg rating: 1.53)
6. B08WJPNZPQ: Lume Natural Deodorant - Underarms and Private Parts - Aluminum-Free, Baking Soda-Free, Hypoallergenic, and Safe For Sensitive Skin - 2.2 Ounce Stick (Twisted Lime) (reviews: 313, avg rating: 2.81)
7. B07PFH92HN: Gel Nail Polish Starter Kit with 36W Nail Light,LED Nail Dryer Lamp 12 Color Gel and Base Top Coat Essentials Manicure Tools (reviews: 232, avg rating: 2.84)
8. B0811S3MZ2: MUSELASH 5 Pairs Upgraded Magnetic Eyelashes Magnetic Eyeliner and Eyelashes Kit,Reusable 3D Magnetic Eyelashes and Strong Magnetic Lashes Set,5 Styles Soft Eyelashes with Professional Tweezers (reviews: 228, avg rating: 2.86)
9. B008C2QYPI: Beauty Town Luxury Wide Band Satin Bonnet Cap (Pink) (reviews: 224, avg rating: 2.95)
10. B00011JN5G: American International Industries Clean + Easy Deluxe Electrolysis Home for Coarse Hair (reviews: 216, avg rating: 2.82)

- Musical Instruments:

1. B0007L8BQW: PYLE-PRO Wireless Clip Lavalier Microphone System Portable Professional Clip Lav lapel Mic set with Volume Control, 20 ft range - Transmitter, Receiver, Battery For Camera, Sound Recorder ProPDWM96 (reviews: 1253, avg rating: 2.72)

2. B09HZMX75B: Uniwit Mini Portable Vocal/Instrument Microphone for Mobile Phone Laptop Notebook Apple iPhone Sumsung Android with Holder Clip (Golden) (reviews: 1193, avg rating: 2.62)
3. B0006DONOK: Emerson EAM-9002 Dual Handheld Wireless Microphone Starter Kit with Wireless FM Receiver, Black (reviews: 391, avg rating: 2.82)
4. B00282WZG0: Grover Jaw Harp (8037) (reviews: 357, avg rating: 2.48)
5. B00H7Y2ZKK: On Stage MN5A Nylon Tip Maple Drum Sticks (Pack of 12) (reviews: 311, avg rating: 2.94)
6. B002UR3JVI: Hello Kitty / Emoji Karaoke Microphone - Style May Vary (reviews: 308, avg rating: 2.55)
7. B01G2DLWZU: Ez-Fret Guitar Attachment, Eliminates Finger Pain, 110 Chords Available, Fits MOST Full Sized Acoustic Guitar, L/H OK, Not A Beginner Tool, For People Whose Fingers Hurt From Guitar Strings (reviews: 299, avg rating: 2.96)
8. B0992FLF8B: 12 Pack Acoustic Panels 1 X 12 X 12 Inches – Acoustic Foam - Studio Foam Wedges - High Density Panels – Soundproof Wedges - Charcoal (reviews: 281, avg rating: 2.95)
9. B06XFYX3NH: Luvay Pocket Guitar, Finger Exercise, Scales & Chords Practice Tool (6 Fret Black) (reviews: 253, avg rating: 2.99)
10. B00CGYETBO: Star MG50-PL Kids Acoustic Toy Guitar 23-Inch, Purple (reviews: 226, avg rating: 2.49)

Common Keywords in Reviews of Products with High Review Count and Low Ratings in category: musical_instruments

- Software:

1. B017250D16: Paramount+ (reviews: 26026, avg rating: 2.68)
2. B00BYJ6BUO: ABC: Watch Live TV & Sports (reviews: 9125, avg rating: 2.93)
3. B01923M8T6: NBC Sports (reviews: 7884, avg rating: 2.77)
4. B01LXOU5PM: NFL (reviews: 6499, avg rating: 2.89)
5. B00E81O27Y: ESPN for Fire TV (reviews: 6185, avg rating: 2.97)

6. B019DCHDZK: fuboTV: Watch Live Sports, TV Shows, Movies & News (reviews: 5437, avg rating: 2.88)
7. B00KY7PEAA: Game of War - Fire Age (reviews: 5433, avg rating: 2.98)
8. B00GY1DWUO: Spectrum TV (reviews: 5306, avg rating: 2.96)
9. B019CVAKMK: Paramount Network (reviews: 4687, avg rating: 2.86)
10. B00BI60DMU: Watch TBS (reviews: 3987, avg rating: 3.00)

- Video Games:

1. B007W5PG5U: SimCity: Limited Edition (reviews: 3765, avg rating: 1.48)
2. B07YBX7Y3P: Fallout 76 - Xbox One Tricentennial Edition (reviews: 3317, avg rating: 2.54)
3. B001FRTK4O: Spore - PC/Mac (reviews: 2400, avg rating: 1.82)
4. B07DK1H3H5: Cyberpunk 2077 - PC [Game Download Code in Box] (reviews: 2277, avg rating: 2.63)
5. B007VTVRFA: SimCity - Standard Edition [Download] (reviews: 2236, avg rating: 1.76)
6. B00ZQB28XK: No Man's Sky - PlayStation 4 (reviews: 1876, avg rating: 2.81)
7. B0B36T478W: ASTRO Gaming A50 Wireless Headset + Base Station Gen 4 - Compatible With PS5, PS4, PC, Mac - Black/Silver (reviews: 1739, avg rating: 2.96)
8. B0BFFJTZNN: SteelSeries Arctis 9X Wireless Gaming Headset – Integrated Xbox Wireless + Bluetooth – 20+ Hour Battery Life – for Xbox One and Series X (reviews: 1731, avg rating: 2.75)
9. B09BXPRNY9: Turtle Beach Recon Chat Xbox Headset for Xbox Series X, Xbox Series S, Xbox One, PS5, PS4, PlayStation, Nintendo Switch, Mobile, & PC with 3.5mm – Glasses Friendly, High-Sensitivity Mic - Black (reviews: 1501, avg rating: 2.83)

10. B003Z0BRDQ: Sid Meier's Civilization V [Download] (reviews: 1490, avg rating: 2.84)

Next are the best sellers of each category. Such products have a high review count and a high average rating score. The top five best selling products are presented in each case.

- Appliances:

  1. 12 Pack Keurig Filter Replacement by K&J - Compatible with Keurig Coffee Machine (2.0 and older) (ASIN: B0B3DB5HTC — Verified Reviews: 11865)
  2. iPartPlusMore Reusable Coffee Filters Compatible with 1.0 and 2.0 Keurig Single Cup Coffee Maker - BPA-Free Stainless Steel Refillable K Cup Coffee Filter with Fine Mesh Screen (Pack of 4) (ASIN: B07RNJY499 — Verified Reviews: 11416)
  3. Linda's Essentials Silicone Stove Gap Covers (2 Pack), Heat Resistant Oven Gap Filler Seals Gaps Between Stovetop and Counter, Easy to Clean (25 Inches, White) (ASIN: B07WTXWC32 — Verified Reviews: 8412)
  4. SAMSUNG Genuine Filters for Refrigerator Water and Ice, Carbon Block Filtration for Clean, Clear Drinking Water, DA29-00020B-3P, 3 Pack (ASIN: B08YBGCNHP — Verified Reviews: 7195)
  5. GE MWF Refrigerator Water Filter — Certified to Reduce Lead, Sulfur, and 50+ Other Impurities — Replace Every 6 Months for Best Results — Pack of 1 (ASIN: B000AST3AK — Verified Reviews: 6854)

- Beauty:

  1. Salux Nylon Japanese Beauty Skin Bath Wash Cloth/towel (3) Blue Yellow and Pink (ASIN: B085BB7B1M — Verified Reviews: 1887)
  2. Godefroy Tint Kit for Spot Coloring, Dark Brown (ASIN: B0BM4GX6TT — Verified Reviews: 1719)
  3. Segbeauty empty bottle 160083 (ASIN: B07C533XCW — Verified Reviews: 1495)

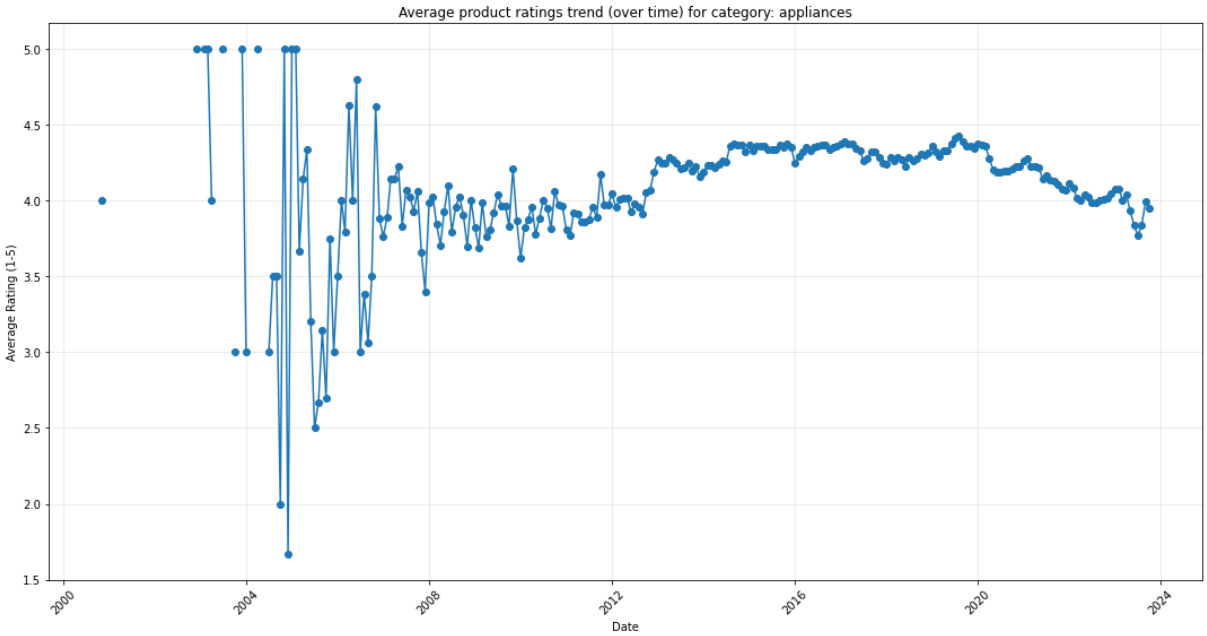
4. Makone Crystal Crowns and Tiaras with Comb Headband for Girl or Women Birthday Party Wedding Prom Bridal Christmas Valentine... (03 Pink) (ASIN: B09X9BG4FC — Verified Reviews: 1317)
5. Collapsible Hair Diffuser by The Curly Co. with The Curly Co. Satisfaction Guarantee (ASIN: B019GBG0IE — Verified Reviews: 1302)

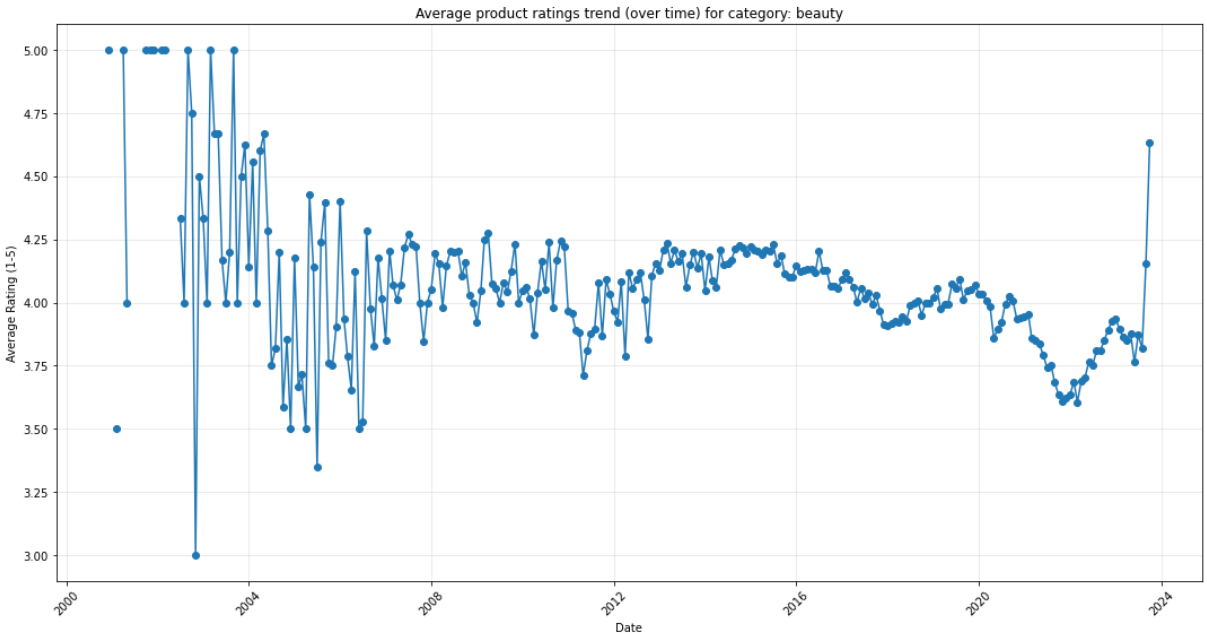
- Musical Instruments:
  1. GLS Audio Instrument Cable - Amp Cord for Bass & Electric Guitar - Straight to Right Angle 1/4 Inch Instrument Cable - Black/Grey Braided Tweed, 10ft (ASIN: B09857JRP2 — Verified Reviews: 9107)
  2. BONAOK Wireless Bluetooth Karaoke Microphone, 3-in-1 Portable Handheld Mic Speaker Machine for All Smartphones, Gifts to Girls Boys Kids Adults All Age Q37(Black Gold) (ASIN: B09W4F2X6S — Verified Reviews: 7030)
  3. Blue Yeti USB Microphone for PC, Mac, Gaming, Recording, Streaming, Podcasting, Studio and Computer Condenser Mic with Blue VO!CE effects, 4 Pickup Patterns, Plug and Play – Blackout (ASIN: B0C6H9T5T6 — Verified Reviews: 6634)
  4. Donner Guitar Delay Pedal, Yellow Fall Analog Delay Guitar Effect Pedal Vintage Delay True Bypass (ASIN: B0BCK6L7S5 — Verified Reviews: 6401)
  5. Neewer Microphone Arm Stand, Suspension Boom Scissor Mic Arm Stand with 3/8" to 5/8" Screw and Cable Ties Compatible with Blue Yeti, Snowball, Yeti X, Quadcast and Other Mics, Max Load 1.5kg (ASIN: B00DY1F2CS — Verified Reviews: 4790)

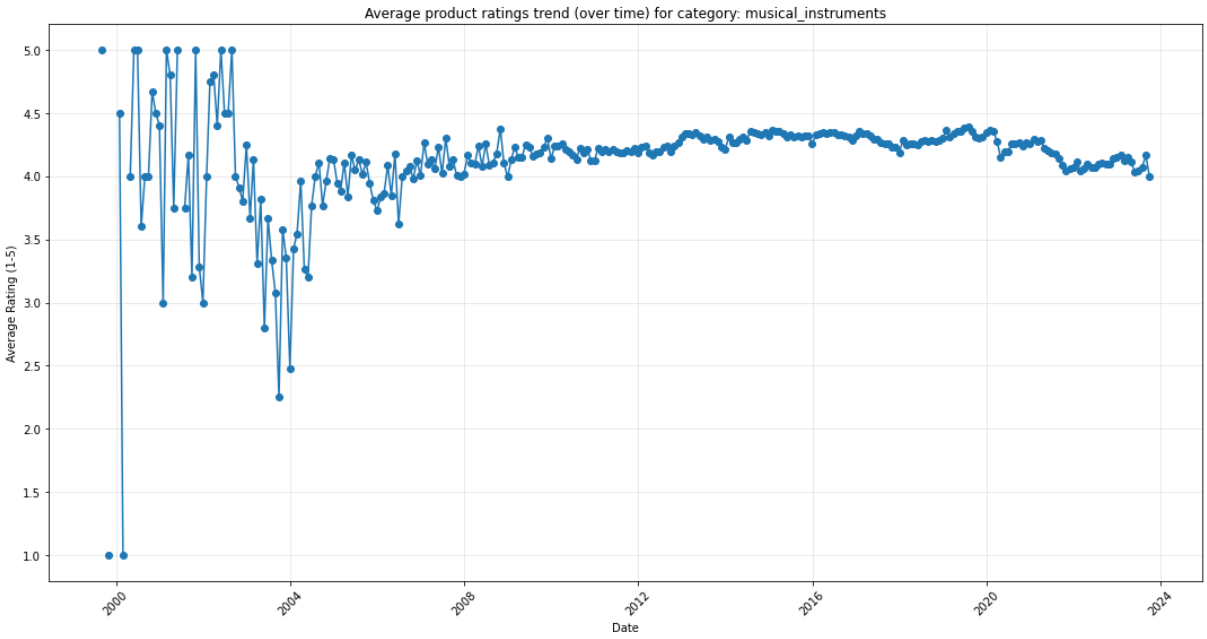
- Software:
  1. Candy Crush Saga (ASIN: B00FAPF5U0 — Verified Reviews: 50276)
  2. Amazon Prime Video (ASIN: B00N28818A — Verified Reviews: 45388)
  3. Minecraft (ASIN: B00992CF6W — Verified Reviews: 42210)
  4. Netflix (ASIN: B005ZXWMUS — Verified Reviews: 32262)
  5. Facebook (ASIN: B0094BB4TW — Verified Reviews: 29836)

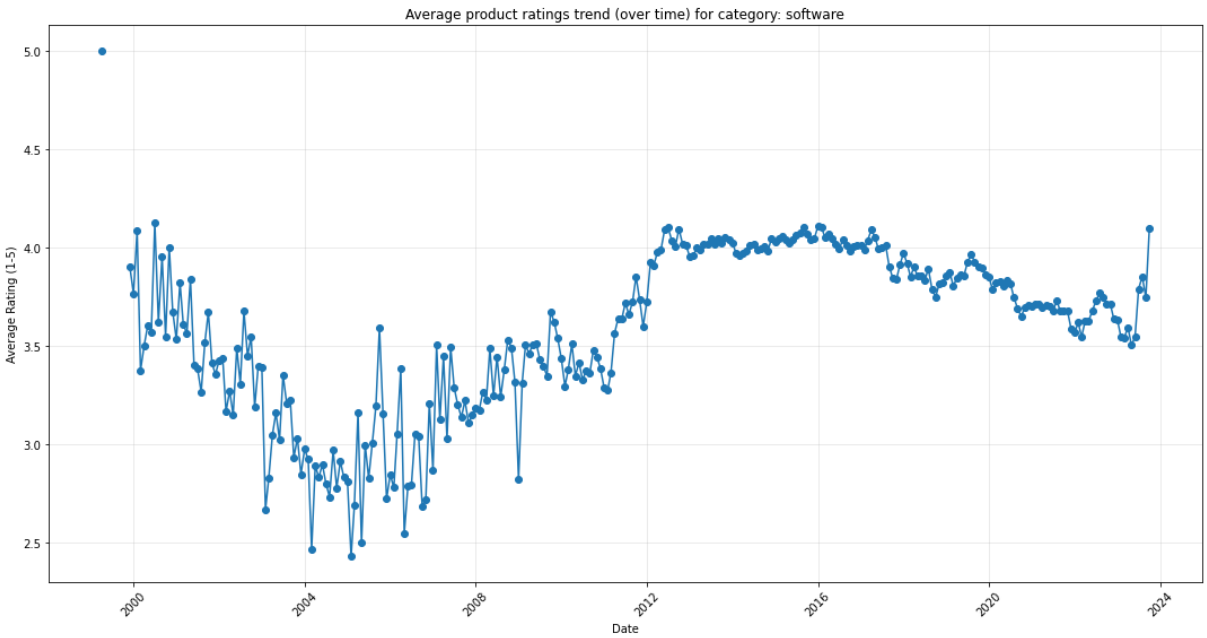
- Video Games:
  1. amFilm Tempered Glass Screen Protector for Nintendo Switch 2017 (2-Pack) (ASIN: B01N3ASPNV — Verified Reviews: 17833)
  2. BENGOO Stereo Pro Gaming Headset for PS4, PC, Xbox One Controller, Noise Cancelling Over Ear Headphones with Mic, LED Light, Bass Surround, Soft Memory Earmuffs for Laptop Mac Wii Accessory Kits (ASIN: B0BN942894 — Verified Reviews: 16403)
  3. DualShock 4 Wireless Controller for PlayStation 4 - Jet Black (ASIN: B077GG9D5D — Verified Reviews: 15195)
  4. Xbox Live Gold: 1 Month Membership [Digital Code] (ASIN: B000N5Z2L4 — Verified Reviews: 12805)
  5. Grand Theft Auto V: Premium Edition - Xbox One [Digital Code] (ASIN: B0086VPUHI — Verified Reviews: 10856)

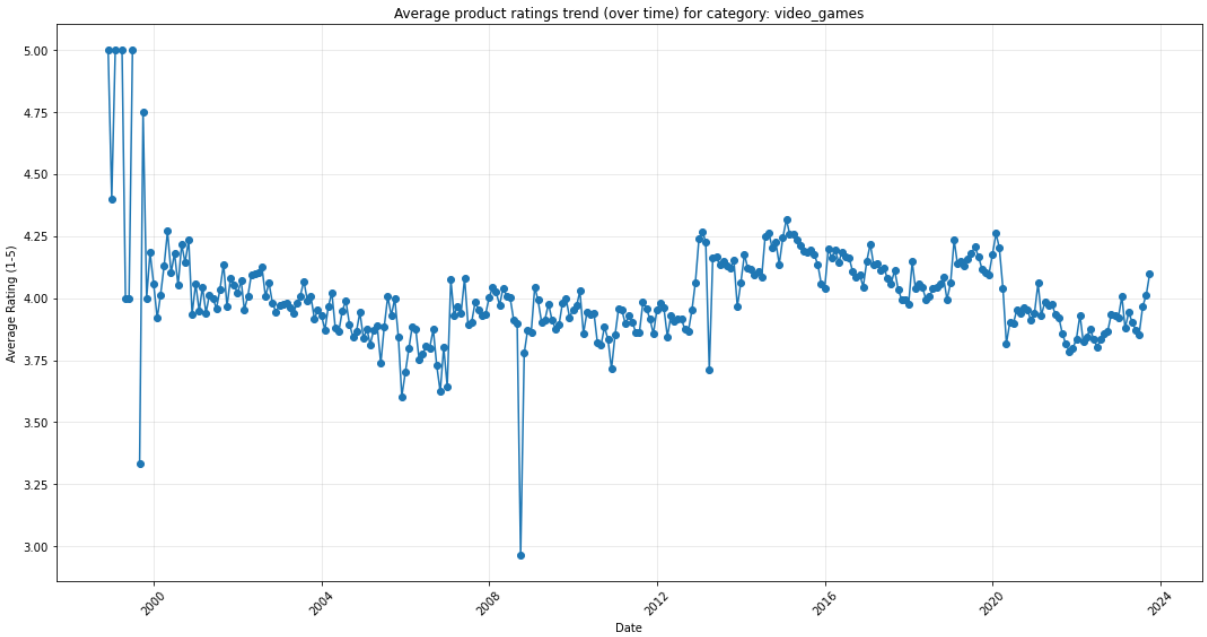
Lastly, the progress of the average product rating throughout the years is shown for each category.






Average product ratings trend (over time) for category: musical_instruments
Average Rating (1-5)
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
2000
2004
2008
2012
2016
2020
2024
Date


Average product ratings trend (over time) for category: software
Average Rating (1-5)
5.0
4.5
4.0
3.5
3.0
2.5
2000
2004
2008
2012
2016
2020
2024
Date



It is evident that there were far too few reviews before 2012 for almost all the categories, as there is an acute fluctuation of the average rating (which is expected, since not many people were using the Internet on a daily basis). However, in almost all categories, there is a point after which the average rating stabilizes around a certain value. This is the common pattern after some years of strong fluctuation due to the smaller number of reviews in the early years.

3.5 Sentiment Analysis Methodology

In this section, an attempt is made to capture the sentiment of all reviews, in order to ultimately compare the textual reviews of products to their respective ratings.

```
def get_sentiment(text):
    return score
```

Listing 1: Sentiment Score Calculation

3.6 Results: Ratings vs. Sentiment

Analysis of the correlation between the numerical rating (1-5 stars) and the calculated sentiment score.

Figure 7: Heatmap of Review Sentiment vs. Ratings

As shown in Figure 7, we observe a strong correlation in the "hot" zones (high rating, high sentiment), indicating that... [Insert your observation about the south-east/north-west consistency here].

4 Part 2: Product Analysis, Clustering and Market Basket Mining

4.1 Introduction to Part 2

Overview of the objectives of Part 2, including product-level text processing, clustering, recommendation logic, and association rule mining.

4.2 Dataset Preparation and Text Pre-Processing

Description of the pre-processing pipeline applied to product metadata and review text.

- Text normalization (lowercasing, punctuation removal)
- Expansion of contractions
- Spell correction
- Stopword removal
- Vectorization using TF–IDF

4.3 Feature Extraction

Explanation of how textual features are transformed into numerical representations suitable for machine learning models.

4.4 Product Clustering

4.4.1 Clustering Methodology

Description of the clustering approach (e.g., K-Means) and distance metrics.

4.4.2 Cluster Selection

Explanation of the elbow method and justification of the chosen number of clusters.

4.4.3 Cluster Interpretation

Qualitative analysis of resulting clusters and their thematic coherence.

4.5 Cluster-Based Recommendation Strategy

Description of the recommendation logic based on cluster membership and product similarity.

4.6 Association Rule Mining

4.7 Temporal Pattern Analysis

Focused analysis on December transactions to capture seasonal purchasing behavior.

5 Conclusion

Summary of the findings from both parts.

- **Part 1 Summary:** Sentiment analysis confirmed that...
- **Part 2 Summary:** Market basket analysis revealed that...